Indian Institute of Management, Rohtak

Global Child Malnutrition Analysis (2025) Targeting Interventions through Machine Learning

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IT Application

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Introduction and Problem Context

The Global Crisis of Child Malnutrition

Childhood malnutrition is not a food scarcity, but a failure of health, economy and infrastructural malfunction. Currently, according to UNICEF, around 45 million children aged below five years are affected by wasting, a phenomenon whereby a child is too skinny in relation to his height. It is a death sentence to millions to whom lack of treatment is an instant death sentence. Nonetheless, the crisis is multi-dimensional. We observe that Stunting (low height relative to age) depicts a state of chronic and long term malnutrition and that Overweight (high weight relative to height) is developing in developing countries due to poor quality of diets rich in calories but low in nutrients.

The Data Science in Humanitarian Aid

The main issue facing international health organizations is not necessarily the absence of data, but the so-called Information Paradox: the problem of excessive data and minimal understanding. There are 154 countries to offer yearly measurements and it is almost impossible to be able to identify using personal hands where the next crisis shall emerge. To fill this gap, this project will use the UNICEF Joint Malnutrition Estimates (JME) 2025 Database. Using the machine learning technique of unsupervised clustering and supervised classification we may shift our reactive model (reacting to a famine) into the proactive model (noticing signs of risk before it makes our world a ruins).

Scope of the Report

The following report presents the pipeline of our work: we are going to start with the processing of the global health indicators and finish with the implementation of a Random Forest model. Instead, we target finding the Crisis Zone, a group of 25 countries, which combined risk of stunting and wasting is the greatest, and examine the socioeconomic factors that went into that group.

Data Methodology and Preprocessing

Source Material

UNICEF JME 2025 Database

The data used is high-resolution survey data of 154 countries. A single entry is a national or sub-national survey. The major indicators examined are five, and they are:

- **Wasted Only (WHZ -2):** Acute malnutrition.
- **Only Stunted (HAZ < -2):** stunted development as a result of chronic malnutrition.
- **Wasted Stunted**: The deadliest of all, signifies both instant and decade-long deprivation.
- **Stunted:** Malnutrition has a sudden impact on the whole child.<|human|>Stunted & Overweight: The double burden of malnutrition.

Malnutrition-Free: The healthy population control.

Technical Preprocessing and Feature Engineering

Developing country data is usually incomprehensive owing to war or absence of infrastructure. In order to prepare it to be used in the Machine learning, we did:

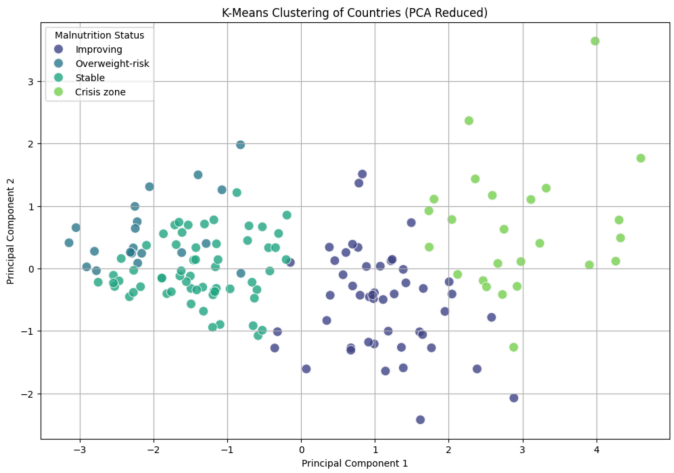
- **Imputation:** Localized mean imputation was used to deal with missing values because global patterns were ignored.
- **Normalization:** The difference in percentages of various indicators has different scales (e.g. is 40 percent stunting and 10 percent wasting) so we used StandardScaler. This ventures to make sure that distances calculated by the K-Means algorithm are in respect to the relative significance of the trend and not the actual significance of the number.
- **Temporal Categorization:** Surveys are made during decades; we categorized them by their year of the survey to see whether the world malnutrition is decreasing in a desired direction or stagnating in the situation in a region such as South Asia (SA) or East and Southern Africa (ESA)

The machine learning framework

Unsupervised Learning

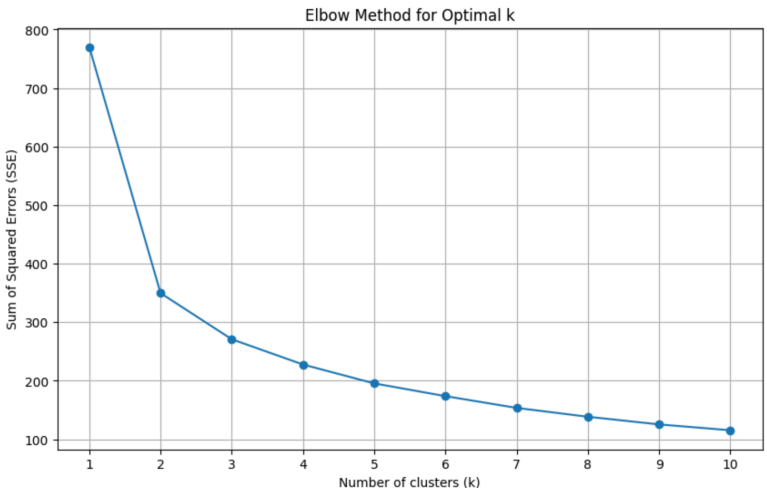
K-Means Clustering

The first objective was to cluster 154 countries into natural ones without any form of bias. We applied the K-Means Clustering, which is a mathematical method of dividing the data points into groups such that the distance between the data points and the cluster centroid is reduced to the minimal.



The Elbow Method

Plotting the Sum of Squared Errors (SSE) we saw the "elbow" at K=4. This validates that the world population of those children who are at Malnourishment naturally takes four unique categories of Severe Risk, Moderate Growth, Stable/Healthy and Nutrition Transition (Overweight risk).



Cluster 1 (the Crisis Zone)

Cluster 1 (the Crisis Zone) engaged stunting and wasting highly thus adding up to a cluster. These are not just starving countries but have undergone biological trauma that does not allow them to grow and lose weight at the same time.

Supervised Learning

Random Forest Classification

After establishing the clusters, we have trained a Random Forest to classify the countries looking for their geographical location and time series characteristics to fall in the Crisis Zone category.

Why Random Forest?

Random Forest unlike a single tree of decision, forms an ensemble of hundreds of trees and votes the majority of the decision. This is paramount to health data, which is usually non-linear and complicated.

Model Training

An 80/20 train-test split was used. The model was tested based on a Confusion Matrix so that we reduce the possibility of a False Negative - the risky error of stating that a country is safe when it is at risk.

Results and Findings Analysis

The Geography of Risk

South Asia and West/Central Africa are the most vulnerable regions as our analysis will confirm. Nonetheless, the statistics indicate a second crisis in a group of the "Nutrition Transition". The wasting declined but at the same time there was a sharp rise of the Stunted and Overweight category in about 20 countries. This is an indicator that these countries are substituting famine with low quality nutrition as they grow resulting in a new form of metabolic health crisis.

Feature Importance Rankings

The Random Forest model found three key drivers that can cause the risk level of a country:

Importance of Survey Year (48.5%)

This points to the fact that risk is very time-sensitive. A nation that was sound five years ago is today under threat with changes of climate or an unstable economy.

Sub-Region (25.2% Importance)

Geography: Geography is a huge determinant of health access.

Wasting/Stunting Ratios

The mathematical correlation between these two variables is the smoking gun of in the long term health fall.

The 25 Crisis Zone Insight

By reducing the number of countries to 25 (154 countries), the model has formed an Insight of a Priority One list. These 25 countries are the most concentrated number of children who will not only experience irreversible cognitive and physical harm but also without intervention.

Economic Impact and Conclusion

The Economic Return on Intervention(ERM)

Malnutrition is no longer a health problem, it is a GDP problem. When such a child is stunted during the first 1,000 days, he or she is likely to earn 20-30 percent as an adult. Our 50-year Return on Investment (ROI) is achieved by targeting aid with Machine Learning in order to provide high-efficiency aid. When a country is able to move out of the Crisis Zone to the Stable Group, the productivity of its workforce has increasingly risen, health care expenses have reduced drastically, and the country will no longer have to rely on foreign assistance.

Conclusion and Future Work

To summarize this analysis, it is important to note that the Sustainable Development Goal 2 (Zero Hunger) of the United Nations can be optimized through the utilization of Machine Learning. K-Means and Random Forest have helped us to develop a reproducible model of at-risk population identification. This model ought to incorporate future climate information (rainfall and crop yield) and conflict indexes in order to have a better focus on the ability to give early warnings.

Cost-Effectiveness

It is highly costly to treat a severely malnourished child in hospital as compared to the cost incurred during early-stage prevention.

ROI in the long term

Stunted children end up earning less in their adulthood. The investment of eliminating malnutrition will be returned within 50 years.

Social Transformation

Intergenerational Health

Mother to malnourished offspring gives birth to malnourished infants; the saving of one life will save the other generation.

Education

Less malnutrition will mean that more children will attend school, will learn and will come out of the poverty trap.

Global Goals

The identified work is directly related to the SDG Goal 2: Zero Hunger.

Recommendations.

Strategic Recommendations

Make the South Asian and West African jump up the queue: Resource allocation should be made urgent to cut stunting and wasting rates in the 2 regions.

Early Warning Systems

This is an exercise that should be implemented to predict countries using the Random Forest predictive model; this will warn them before they deteriorate

Sustainable Development

Attention should be paid to the maintenance of the health system in the states where they are Stable, and special attention should be paid to the reaction on the special problem of the Nourishment Transition in the overweight-threat countries.

References

- UNICEF Joint Malnutrition Estimates (JME) 2025 Dataset.
- United Nations Sustainable Development Goal 2 (Zero Hunger).
- Child Malnutrition Analysis Methodology (K-Means & Random Forest).